

Problem 7.15

A treasury bill or T-Bill with face value \$100 is a security which is exchangeable for \$100 on the maturity date. Suppose that a T-Bill with face value \$100 is issued on 09/09/2005 and matures on 03/09/2006 (there are 182 days between the dates).

Assuming simple interest, given that the annual rate of interest is 2.810%, find the price of the security.

Solution

Let P be the price of the T-Bill at issue. Under simple interest,

$$100 = P(1 + it),$$

where

$$i = 0.02810, \quad t = \frac{182}{365}.$$

Thus,

$$100 = P \left(1 + 0.02810 \cdot \frac{182}{365} \right).$$

Solving for P ,

$$P = \frac{100}{1 + 0.02810 \left(\frac{182}{365} \right)}.$$

Evaluating,

$$P \approx 98.6182.$$

Therefore, the price of the security is

$$\boxed{\$98.6182}.$$